

Regular/Supplementary Winter Examination – 2023

Semester : V

Duration: 3 Hr.

1. All the questions are compulsory.
2. Use of non-programmable scientific calculators is allowed.
3. Assume suitable data wherever necessary and mention it clearly.

	Marks
Q. 1 Solve Any Two of the following.	12
A) Define α and β for BJT and develop a relation between the two. Why is α less than 1 and β more than 1?	6
B) Compare Power MOSFET with BJTs.	6
C) A BJT with current gain $\beta = 50$, has load resistance 10Ω , dc supply voltage $V_{cc} = 120V$ and input voltage to base circuit, $V_B = 10V$, for $V_{CES} = 1.2V$ and $V_{BES} = 1.6V$. Calculate,	6
a) R_B For operation in saturated state.	b) R_B for ODF 6
c) β_f	d) power loss for both parts (a) and (b)
Q.2 Solve Any Two of the following.	12
A) Explain single phase half wave controlled rectifier of R load with circuit diagram, waveforms and necessary equations.	6
B) Describe a single phase full wave midpoint controlled rectifier with R load with relevant waveforms and derive expression for average and rms values of output voltage.	6
C) A load of 10Ω is connected through half wave controlled rectifier to 220V, 50Hz, single phase source. Calculate the power delivered to load for a firing angle of 60° . Find also the value of input power factor.	6
Q. 3 Solve Any Two of the following.	12
A) Explain different types of control strategies of chopper.	6
B) Explain steady state analysis of Class A chopper.	6
C) Explain buck-boost converter with circuit diagram, equations and necessary waveforms.	6
Q.4 Solve Any Two of the following.	12
A) Draw and explain single phase full bridge inverter with necessary waveforms.	6
B) Draw and explain 3 phase inverter with 180° mode. Draw phase and line voltage waveforms.	6
C) Explain PWM technique and its 3 types in detail.	6

- Q. 5 Solve Any Two of the following.** **12**
- A) Draw and explain single phase half wave AC voltage controller with R Load. **6**
- B) Explain step up and step down cyclo-converter with necessary waveforms. **6**
- C) Explain 3 phase to 1 phase cyclo-converter with necessary waveforms. **6**

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